

DEPARTMENT OF CHEMISTRY

GIRIJANANDA CHOWDHURY UNIVERSITY, ASSAM

B.Tech. 2nd Semester Practical Examination, 2025

Subject : Chemistry Lab

Subject Code : BCH23111P

Time : 3 Hours

Marks = 30

- Q.1. Estimate the hardness of the given water sample by a standard solution of EDTA. (15)
Q.2. Viva – voce. (10)
Q.3. Practical Notebook (5)

Procedure

(A) Prepare **100 mL** of a standard (0.1M) solution of CaCO_3 and write the whole process showing relevant calculations.

(B) Standardization of the EDTA solution :-

Pipette out 10 ml of the standard Ca^{2+} solution into a conical flask. Add to it 2 mL of $\text{NH}_3 - \text{NH}_4\text{Cl}$ buffer solution followed by 30 mg of solochrome black-T indicator, turning the solution wine red. Titrate this solution against the EDTA solution (taken in a burette) till the colour of the solution changes from wine red to pure blue, which is the end point. Repeat the titration to obtain at least two concordant readings.

(C) Estimation of hardness of the water sample:-

Pipette out 50 ml of the water sample into a conical flask. Add to it 2 mL of $\text{NH}_3 - \text{NH}_4\text{Cl}$ buffer solution followed by 30 mg of solochrome black-T indicator, turning the solution wine red. Titrate this solution against the standardized EDTA solution till the colour of the solution changes from wine red to pure blue, which is the end point. Repeat the titration to obtain at least two concordant readings.

(D) Calculation.

(E) Conclusion.

N.B. For procedure (B) & (C) described above must not be written in the answer copy. Only the relevant chemical equations, stoichiometric deductions and tables need to be written.